DIPLOMA PROGRAMME PRACTICE / HL / SET B

Economics / Paper 3

Question booklet · 105 minutes · 60 marks

Answer all parts of both questions. A calculator is permitted. Write your answers and diagrams in the spaces provided. Show working and units. Give exact values or round to two decimal places unless otherwise instructed.

QUESTION 1 / 30 MARKS

A tariff on an imported staple

Lumen is a small price-taking economy. With free trade, grain costs 12 dinars per sack. Domestic demand Qd = 200 − 5P and supply Qs = 20 + 5P, measured in thousands of sacks each month. Traders can import freely at the given world price. Assume competitive markets and no transport costs or externalities for the calculations.

The farm ministry proposes a tariff of 3 dinars per sack. Farmers say volatile earnings discourage investment. In the poorest fifth of households grain accounts for 18% of expenditure; in the richest fifth it accounts for 3%. A trial of crop storage reduced post-harvest losses but requires reliable electricity. The government can alternatively fund time-limited income support for small farmers from general taxation.

a(i) Calculate the free-trade quantities supplied and demanded domestically. **[2]**

a(ii) Calculate monthly spending on grain imports under free trade. **[2]**

a(iii) Calculate the level of imports after the tariff and the resulting monthly tariff revenue. **[3]**

a(iv) Calculate the loss of consumer surplus per month caused by the tariff. **[3]**

a(v) Calculate the gain in domestic producer surplus per month. **[2]**

a(vi) Calculate the change in national welfare per month. Explain why counting only tariff revenue gives a misleading result. **[4]**

a(vii) Using a tariff diagram, distinguish the part of the efficiency loss due to excess domestic production from that due to reduced consumption. **[4]**

b Using the information and your knowledge of economics, recommend a policy to support vulnerable grain farmers while protecting access to food in Lumen. **[10]**

QUESTION 2 / 30 MARKS

A currency under pressure

Meridia's central bank maintains a managed exchange rate. The rate begins at 5 local units per US dollar; maintaining the announced rate requires sales of foreign reserves when demand for dollars rises. Fuel and medicine are imported under short contracts. Some exporters can increase shipments after a year, but transport capacity is currently tight. Local firms service unhedged dollar loans from local-currency revenues.

A possible new rate is 6 local units per US dollar. All flows in the table are annual US-dollar billions. Ignore valuation changes and errors for the flow calculation. Treat net private financing as the only non-reserve funding for the current-account balance. Beginning reserves are US$6 billion. Annual imports of goods and services used in the reserve-cover calculation are US$12 billion.

Current-account item	Receipts	Payments
Goods	8.0	10.0
Services	3.0	2.0
Primary income	0.4	1.3
Secondary income	0.7	0.4

a(i) Calculate the goods balance and the services balance. **[2]**

a(ii) Calculate the current-account balance. **[2]**

a(iii) Net private financing supplies US$1.0 billion. Calculate reserve use, ending reserves and months of import cover. **[3]**

a(iv) Calculate the percentage rise in the local-currency price of a US dollar and the percentage fall in the dollar value of a local unit if the new rate is adopted. **[3]**

a(v) A firm owes US$4 million due immediately. Calculate the change in the local-currency value of this liability at the new rate. **[2]**

a(vi) Using a foreign exchange diagram for US dollars, explain how intervention can maintain the initial exchange rate when demand for dollars rises. **[4]**

a(vii) Explain why lowering the value of the local currency might initially widen the trade deficit measured in local currency. **[4]**

b Using the information and your knowledge of economics, recommend an exchange-rate policy for Meridia in response to the continuing demand for dollars. **[10]**